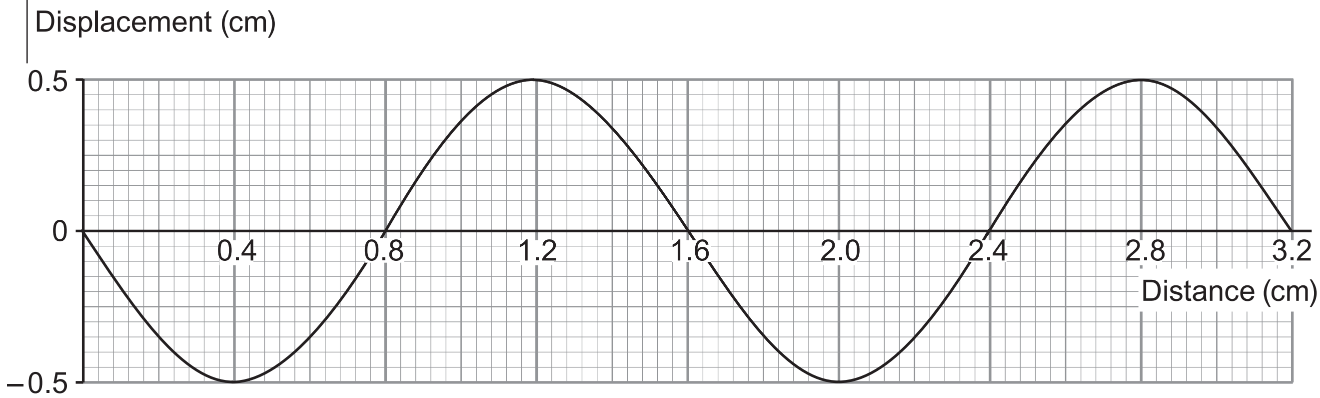
Examiner only

Answer **all** *questions.*

1. A teacher demonstrates water waves in a ripple tank. The diagram represents the waves.



(a) (i) State the wavelength of the wave. .... cm [1]

(ii) State the amplitude of the wave. .... cm [1]

(b) It is observed that 10 waves pass a point in the tank in 2 s.

(i) <u>Underline</u> the correct value for the frequency of the waves.

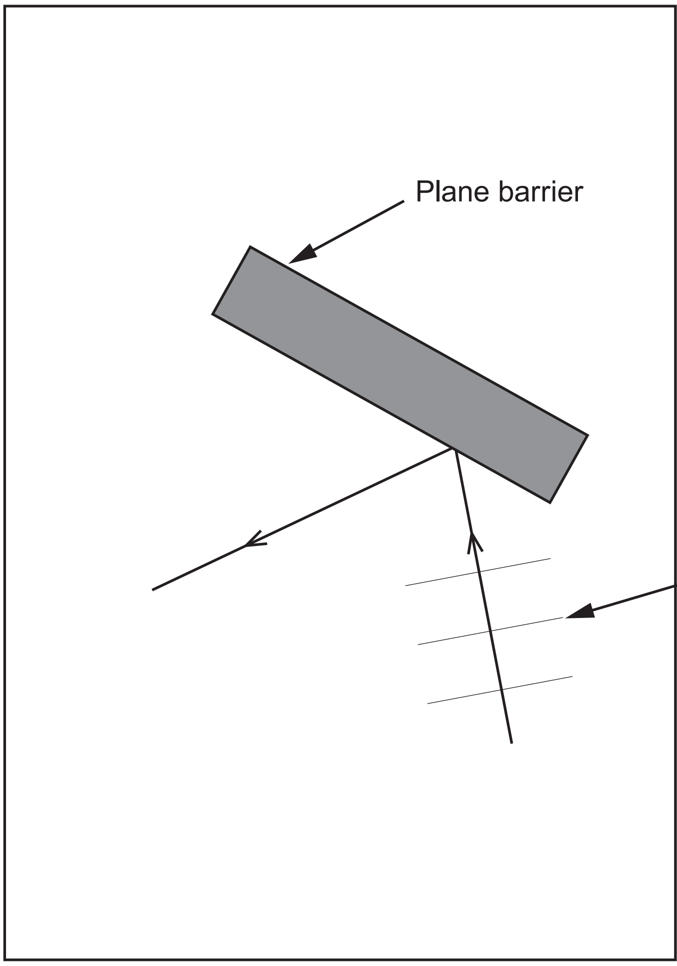
2 Hz 5 Hz 10 Hz 20 Hz [1]

(ii) Select an equation from page 2 and use it to calculate the wave speed. [3]

Wave speed = ... cm / s

Examiner only

(c) The waves are made to reflect off a plane barrier. **Complete the diagram** to show the reflected wavefronts. [2]



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2. Electromagnetic (em) waves are widely used in a variety of ways. Satellites rely on microwaves to communicate with Earth as the microwaves penetrate through the upper layers of our atmosphere. Infra-red waves have a higher frequency than microwaves and are used widely in communications. X-rays and gamma rays are ionising radiation and are both used in medicine; gamma rays can be used in cancer treatment, shrinking tumours. Radio waves are used to transmit TV and radio signals.

(a) The diagram shows an incomplete em spectrum. **Complete the diagram** to show the em spectrum in order. [2]

Radio waves	....	....	Visible light	UV light	....	Gamma rays

(b) All em waves are transverse. Describe what is meant by a transverse wave. [2]

....

....

....

(c) Tick (✓) the boxes alongside the **two** correct statements about em waves below. [2]

Only radio waves and microwaves transmit information ☐

All em waves travel at the same speed in a vacuum ☐

All em waves are a form of radiation ☐

All em waves have the same frequency ☐

Radio waves have the smallest wavelength ☐

(d) A student suggests that radio waves are just as harmful to humans as gamma rays. Explain whether or not you agree with this statement. [2]

....

....

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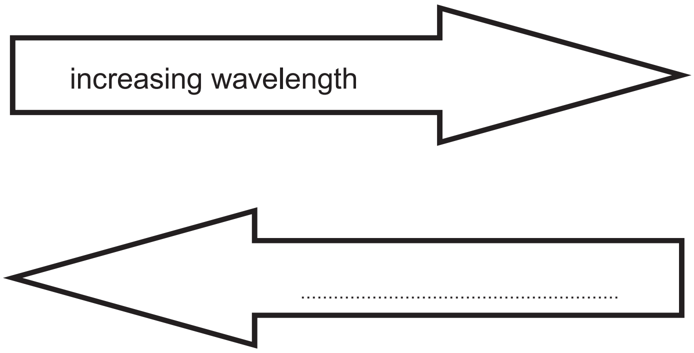
8

Examiner only

Answer **all** *questions.*

1. Various types of radiation form a continuous spectrum called the electromagnetic (em) spectrum. The diagram shows the regions of the em spectrum.

Gamma rays	X-rays	Ultraviolet	Visible light	**A**	Microwaves	Radio waves



(a) (i) Complete the arrow above, using **only one** of the following phrases. [1]

- **decreasing speed**
- **increasing speed**
- **increasing frequency**
- **decreasing energy**

(ii) Name the em radiation labelled **A** in the diagram. [1]

....

(b) (i) Name the ionising em wave given out by some radioactive materials. [1]

....

(ii) State why ionising em radiation is dangerous to humans. [1]

....

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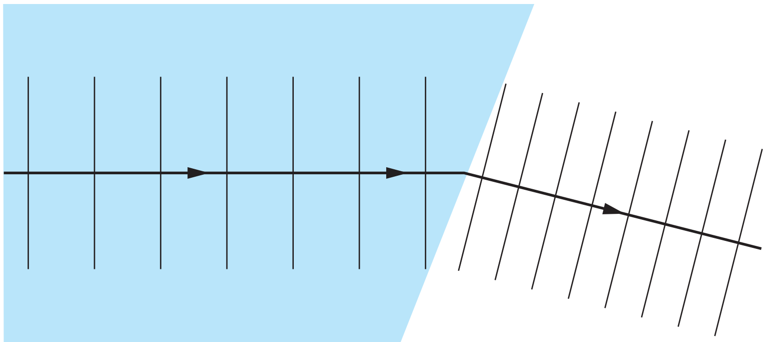
Examiner only

3. A ripple tank is used to demonstrate what happens to water waves as they travel from deep water into shallow water.

The results are shown in the diagram below.



(a) The water waves change direction at the boundary between deep and shallow water.

State the name of this effect. .... [1]

(b) Tick (✓) the boxes next to the **two** correct statements that describe the water waves. [2]

The frequency of the waves is higher in shallow water	☐
The amplitude of the waves is the same in deep and shallow water	☐
The wave speed is less in shallow water	☐
The wavelength decreases as the waves pass from deep to shallow water	☐

(c) In deep water the waves have a wavelength of 3 cm and a frequency of 5 Hz. Use this information and an equation from page 2 to calculate the wave speed. [2]

Wave speed = ... cm/s

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2. The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays

(a) **Tick** (✓) the **three** correct statements. [3]

The frequency of em waves decreases from radio waves to gamma rays. ☐

All em waves travel at the same speed in a vacuum. ☐

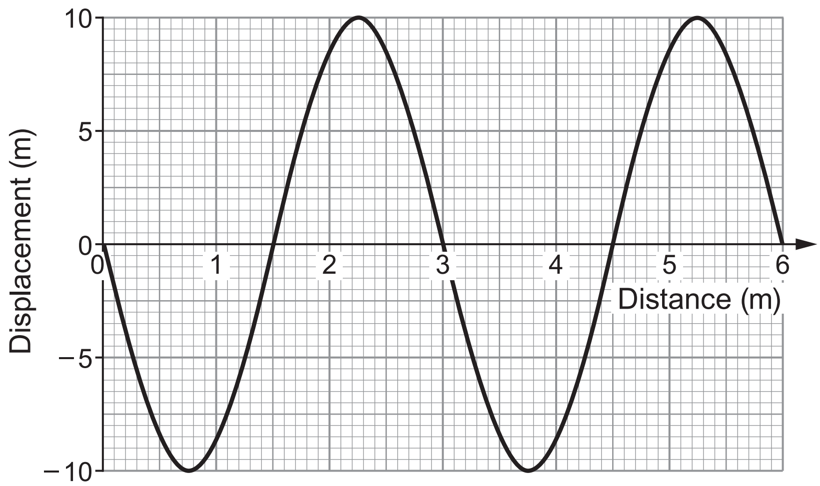
Gamma rays can be emitted from radioactive materials. ☐

The wavelength of em waves decreases from radio waves to gamma rays. ☐

Radio waves, microwaves, infra-red and visible light are all ionising radiations. ☐

All em waves are longitudinal. ☐

(b) The graph below shows a wave diagram.



Use the following values to answer the questions below. You may use each value once, more than once or not at all.

2	3	5	6	10

(i) I. State the amplitude of the wave. .... m [1]

II. State the number of complete waves shown. .... waves [1]

III. State the wavelength of the wave. .... m [1]

Examiner only

(ii) I. Complete the following sentences by <u>underlining</u> the correct phrase in each bracket. [2]

The wave has a frequency of (**7.5 Hz / 7.5 cm/s / 7.5 cm**).

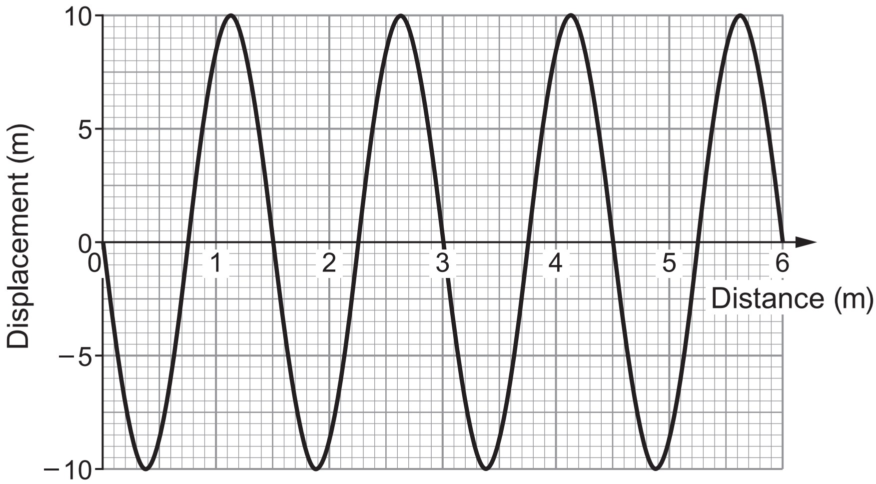
This means 7.5 waves pass a point each (**second / minute / hour**).

II. Use information from parts (b)(i) and (ii) to calculate the wave speed.
Use the equation:

wave speed = wavelength × frequency [2]

Wave speed = ... m/s

(c) The speed of the wave remains constant but its **frequency is doubled**.
A student draws a diagram of the expected wave. It is shown below.



Explain whether you agree with the student's diagram of the wave. [2]

....

....

....

....

12

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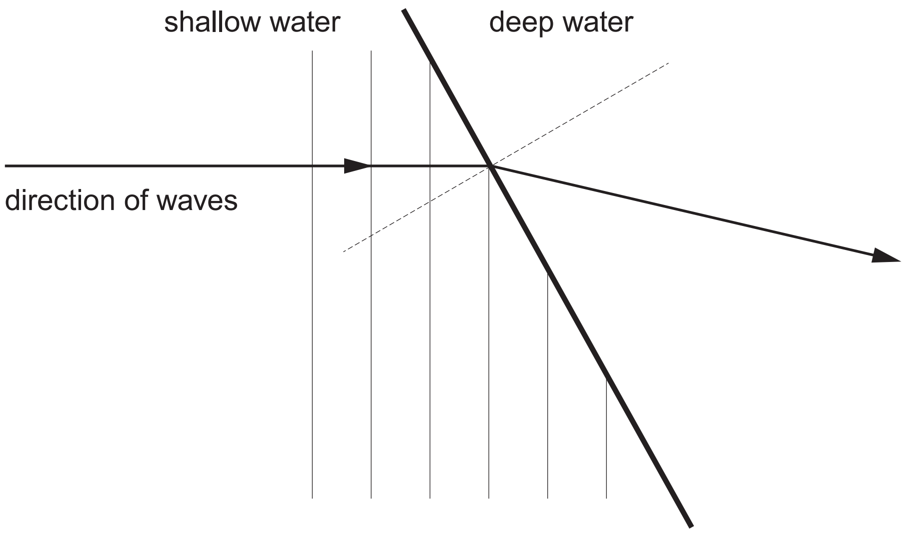
Examiner only

7. Students study refraction of waves in a ripple tank.

They set up the tank to show waves going from shallow water into deeper water.

The diagram shows a number of wavefronts approaching a boundary between shallow and deep water.

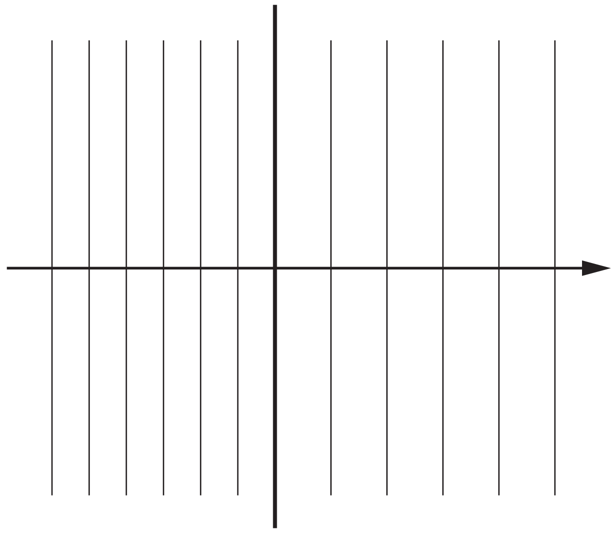
The first four wavefronts have reached the boundary.



(a) **Complete the diagram** to show the wavefronts in the deep water. [3]

Examiner only

(b) In a separate experiment, the wavefronts pass over the boundary in the way shown below.



Sam says that the speed of the waves is greater in deep water than shallow water.

Explain whether you agree with his statement. [3]

....

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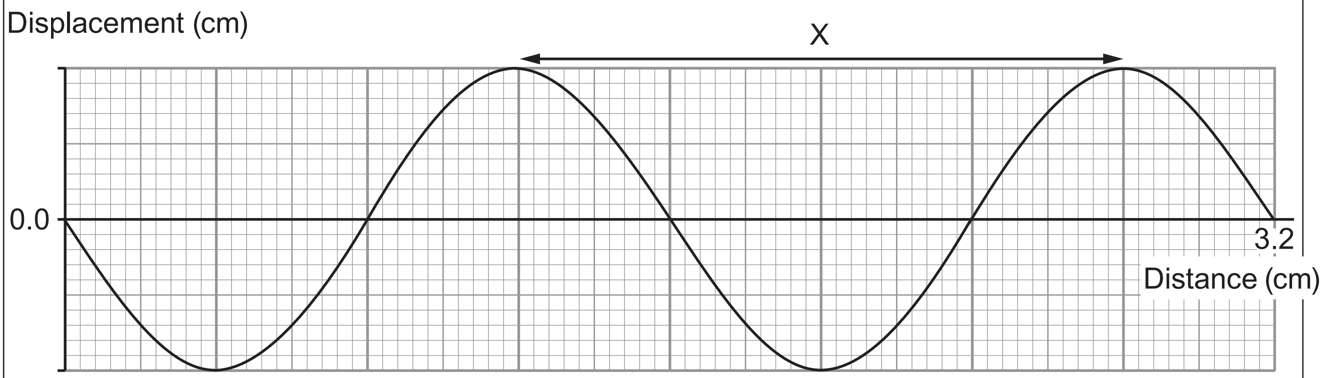
END OF PAPER

Examiner only

2. A teacher demonstrates water waves in a ripple tank.

(a) The diagram represents the waves produced at a particular frequency.



(i) Use the information in the diagram to calculate the distance X. [2]

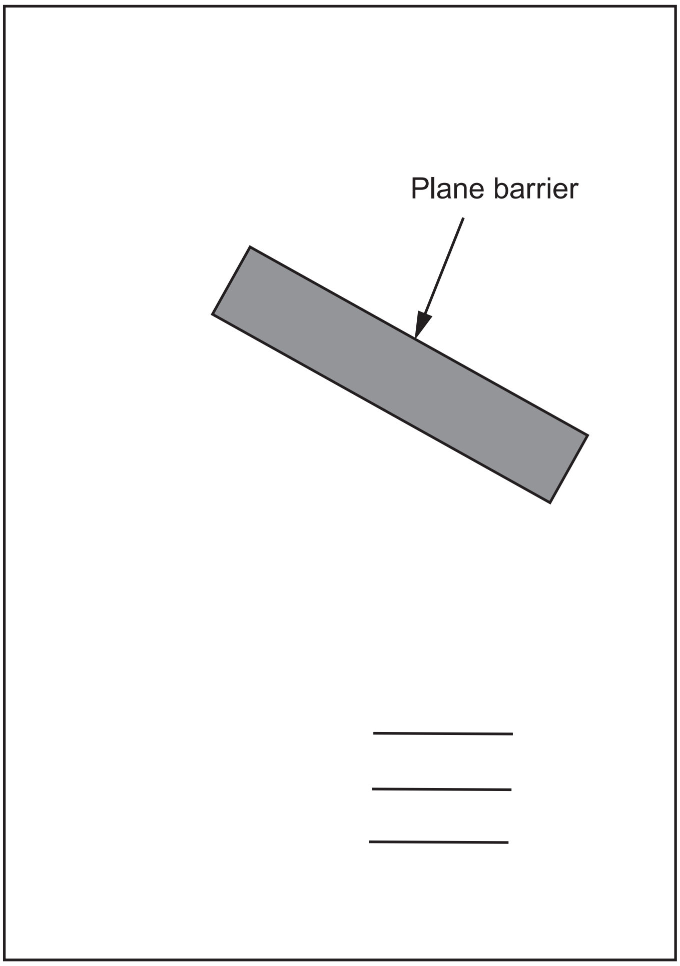
X = ... cm

(ii) A wave takes 5 s to travel 40 cm along the surface of the water in the ripple tank. Use equations from page 2 and the information above to calculate the frequency of the wave. [4]

Frequency = ... Hz

Examiner only

(b) Wavefronts travel towards a plane barrier. **Complete the diagram** to show the path of the wavefronts as they are incident on and reflect off the barrier. [2]



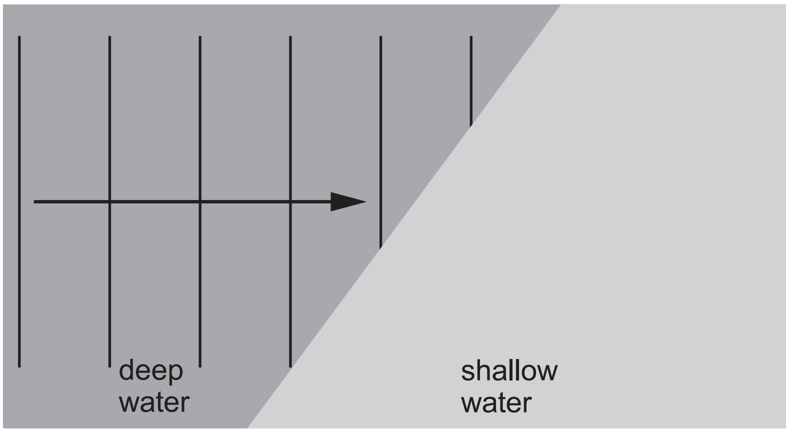
3430UC01 07

8

Examiner only

2. A physics teacher demonstrates the refraction of water waves in a ripple tank.

(a) **Complete the diagram below** to show the wavefronts in the shallow water. [2]



(b) The wave generator in the deep water produces 5 waves in 10 s. Use an equation from page 2 and a measurement from the diagram to determine the speed of the waves in the deep water. Give your answer in m/s. [3]

Wave speed = ... m/s.

(c) State how the frequency, wavelength and wave speed will compare in deep and shallow water. [3]

....

....

....

....

....

....

8

Examiner only

6. Satellites in orbit around the Earth are used for a variety of purposes. One use is communications between base stations and satellites. Satellites for communications are usually put into geostationary orbits, orbiting above the equator. Satellites for other uses such as weather forecasting can be put in polar orbits. The speed of the satellite and therefore its orbital period, the time for one orbit of the Earth, is determined by its orbit height.

(a) Another kind of orbit is a geosynchronous orbit. State **one** similarity and **one** difference between a geostationary and a geosynchronous orbit. [2]

....

....

....

....

(b) The table below gives data on the orbits of different satellites.

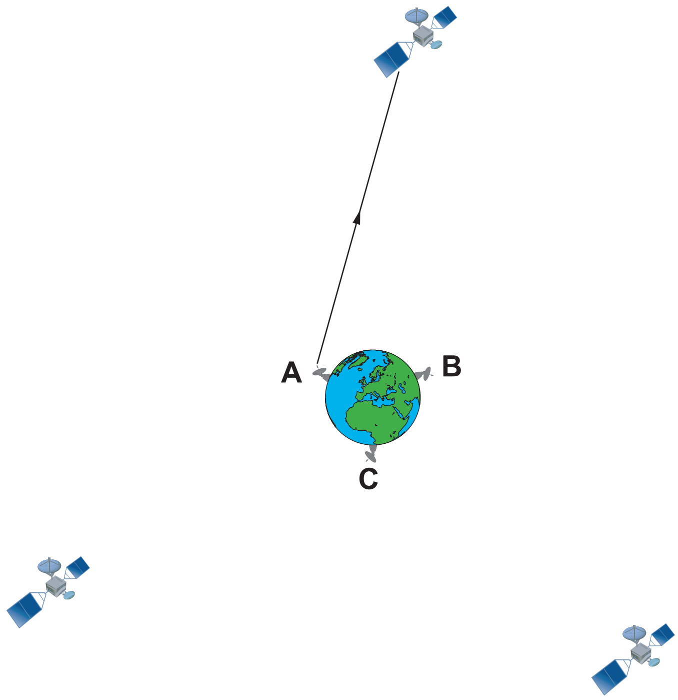
Satellite	Height above Earth (km)	Period (hours)
1	21450	12.9
2	24000	15
3	36000	24
4	507	1.6
5	1090	1.8
6	20200	12

(i) State which satellite is in geostationary orbit. [1]

....

Examiner only

(ii) A signal is sent from base station **A** to a geostationary satellite and is received at another base station after a time delay of 0.48 s. An incomplete diagram of the signal's path is shown.



Use an equation from page 2 to calculate the distance the signal has travelled. The speed of light, $c = 3 \times 10^8$ m/s. [3]

Distance = ... m

(iii) Use your answer to (ii) and the information in the table to determine which base station, **A**, **B** or **C**, finally receives the signal. Show all your workings. [2]

Receiving base station = ...

8

END OF PAPER

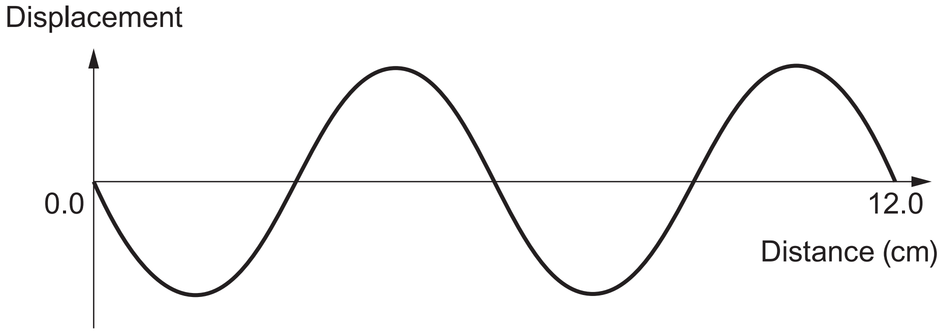
 (3430UC0-1)

Turn over.

Examiner only

2. Microwaves travel through space at a speed of 3×10^8 m/s and are one part of the electromagnetic spectrum.

(a) The diagram represents microwaves produced at a certain frequency.



(i) Use the information in the diagram to calculate the wavelength of these microwaves. [1]

Wavelength = ... cm

(ii) Use your answer above and the equation:

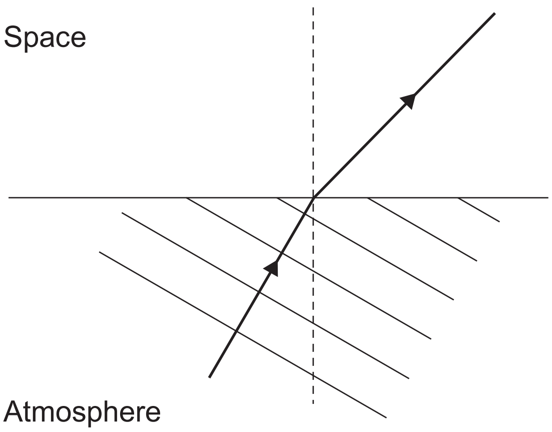
wave speed = wavelength × frequency

to calculate the frequency of these microwaves. [3]

Frequency = ... Hz

Examiner only

(b) Microwaves are refracted as they travel from the atmosphere into space.



Complete the diagram to show the refracted wavefronts in space. [3]

(c) Geostationary artificial satellites are used to send microwave signals around the world.

(i) Explain why a satellite in a geostationary orbit above the equator appears to stay in a fixed place above the surface of the Earth even though both the satellite and the Earth are constantly moving. [2]

....

....

....

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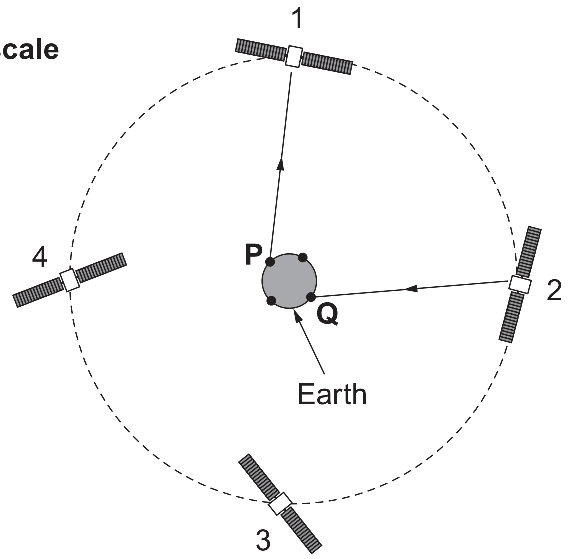
Examiner only

(ii) The diagram shows 4 satellites in geostationary orbit around the Earth.
A signal is to be sent from Earth station **P** to **Q** using the satellites 1 and 2.
The first and last parts of the path are shown.

Diagram not to scale



Use an equation from page 2 to calculate the time taken for the signal to travel from **P** to **Q**.
The height of a geostationary satellite above the Earth is 3.6×10^7 m. [3]
(Speed of light, $c = 3 \times 10^8$ m/s)

Time = ... s

(iii) It was suggested that if the signal had been sent in the opposite direction from **Q** to **P** via satellites **3** and **4** then it would have arrived quicker.
Explain whether you agree with this suggestion. [1]

....

....

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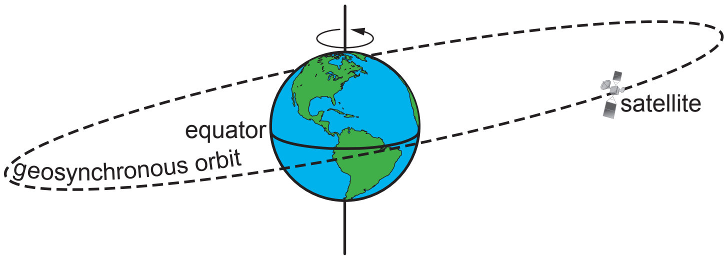
Examiner only

5. The diagram shows an example of a satellite in geosynchronous orbit around the Earth.

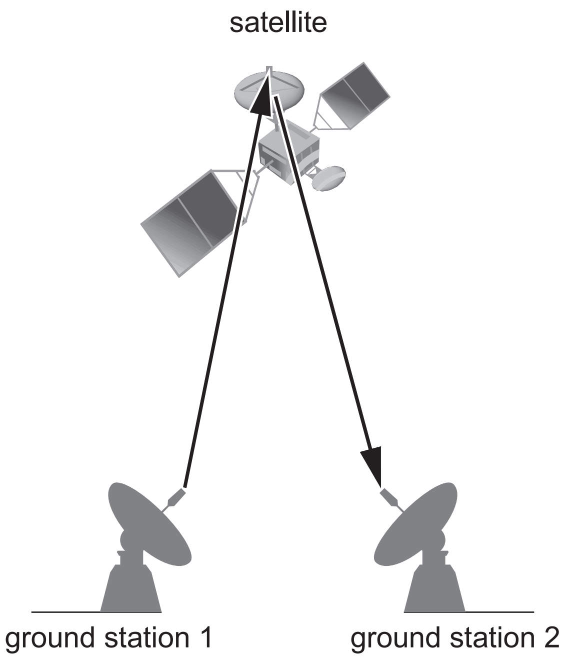


(a) State **one** way in which a geostationary orbit is different to the one shown above. [1]

....

....

(b) The diagram shows a signal sent from one ground station to a satellite which is then received back at a second ground station.



Examiner only

(i) Use the equation:

$$\text{time} = \frac{\text{distance}}{\text{speed}}$$

to calculate the time interval between a microwave signal being sent from Earth to one of the satellites 36 000 000 m away and it being received back again. [3]
(speed of light, $c = 3 \times 10^8$ m/s)

time = ... s

(ii) Microwaves travel at the speed of light and have a range of wavelengths between 0.002 m and 1 m.
Use an equation from page 2 to calculate the **maximum** frequency of microwave signals. [3]

maximum frequency = ... Hz

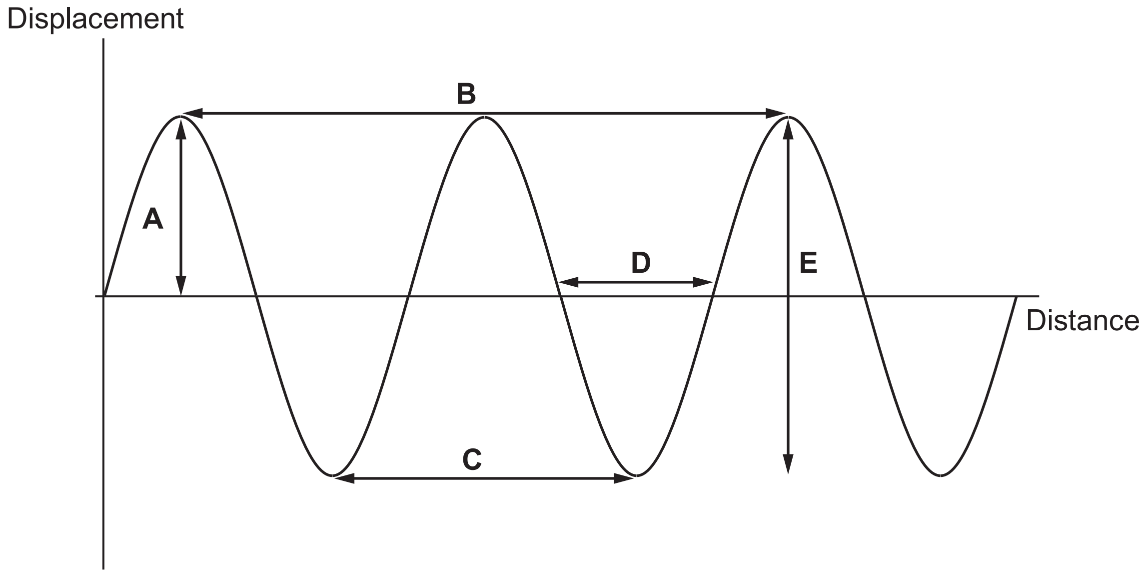
(iii) Name **one** type of electromagnetic radiation which has frequencies smaller than those of microwaves. [1]

....

8

Examiner only

2. The diagram represents water waves on the surface of a swimming pool.



(a) How many complete waves are shown in the diagram? .... [1]

(b) Which letter **A**, **B**, **C**, **D** or **E** shows the amplitude of the wave? .... [1]

(c) Select a word from the box to complete the following sentences.

frequency	time	speed	wavelength	amplitude

(i) The number of waves per second is the .... [1]

(ii) The maximum displacement of the wave is the .... [1]

(d) A wave peak takes 0.5 s to travel the distance labelled **B** on the diagram. The speed of the wave is 20 cm/s.

(i) Use the equation:

$$\text{distance} = \text{speed} \times \text{time}$$

to calculate the distance travelled by the wave. [2]

Distance = ... cm

(ii) Calculate the wavelength. [1]

Wavelength = ... cm

Examiner only

(e) Water waves are transverse waves.

(i) Give another example of a transverse wave. [1]

....

(ii) Sound waves are not transverse waves. What type of waves are they? [1]

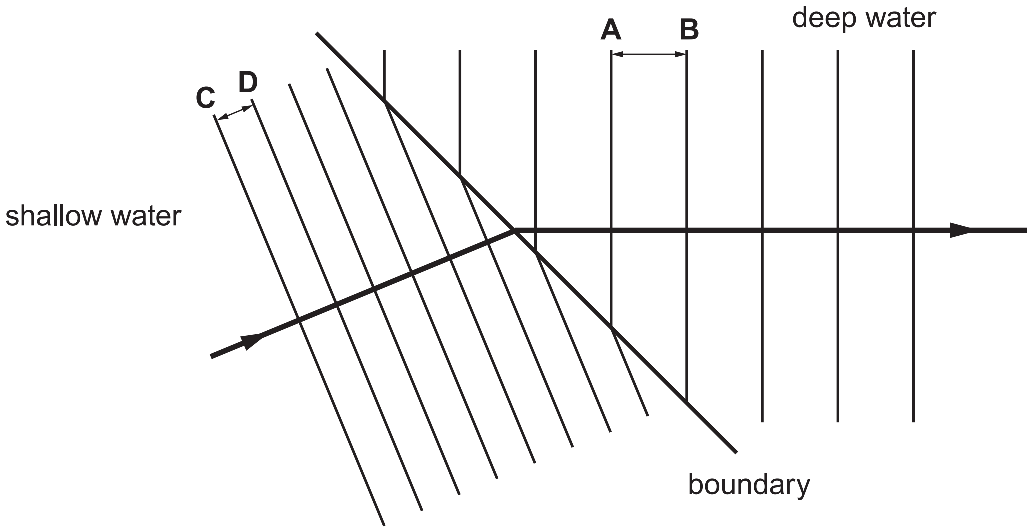
....

9

3420U101 07

Examiner only

8. In class, a teacher demonstrates refraction using a ripple tank. The diagram below shows plane wavefronts travelling across a boundary between shallow and deep water. The frequency of the waves remains **constant** during refraction.



(a) Using a ruler, students measure the distance between wavefronts **A** and **B**. This measurement is the wavelength of the water waves in deep water. The distance between wavefronts **C** and **D** is measured to obtain their wavelength in the shallow water. The results are shown below.

	Deep water (**AB**)	Shallow water (**CD**)
Wavelength (mm)	10	5

(i) State how the measurement of wavelength could be improved. [1]

....

....

(ii) The wavelength in the deep water is twice the wavelength in the shallow water. The teacher suggests, “the speed of the wavefronts in shallow water is double the speed of the wavefronts in the deep water.” Using information provided explain if the suggestion made by the teacher is correct. [2]

....

....

....

Examiner only

2. The diagram shows regions of the electromagnetic (em) spectrum.
The regions are in order.

radio waves	microwaves	infra-red	visible light	ultraviolet	X-rays	gamma rays

(a) **Tick** (✓) the **three** correct statements. [3]

The frequency of em waves decreases from radio waves to gamma rays. ☐

All em waves travel at the same speed in a vacuum. ☐

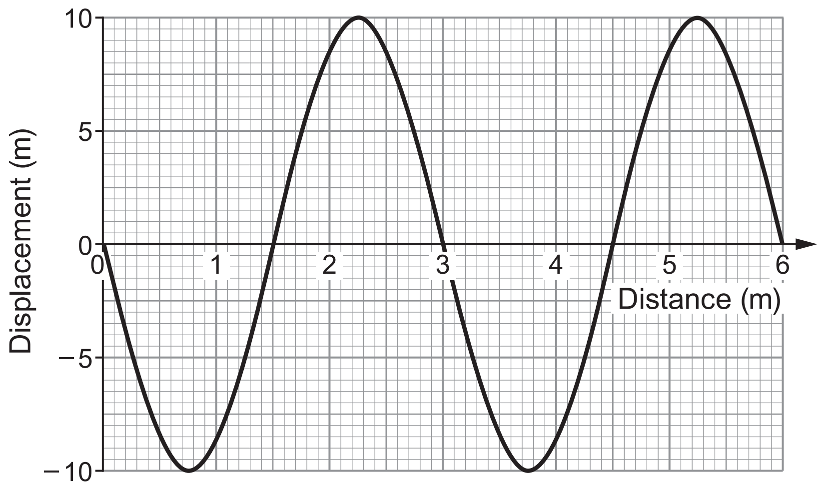
Gamma rays can be emitted from radioactive materials. ☐

The wavelength of em waves decreases from radio waves to gamma rays. ☐

Radio waves, microwaves, infra-red and visible light are all ionising radiations. ☐

All em waves are longitudinal. ☐

(b) The graph below shows a wave diagram.



Use the following values to answer the questions below. You may use each value once, more than once or not at all.

2	3	5	6	10

(i) I. State the amplitude of the wave. .... m [1]

II. State the number of complete waves shown. .... waves [1]

III. State the wavelength of the wave. .... m [1]

(3420U10-1)

Examiner only

(ii) I. Complete the following sentences by underlining the correct phrase in each bracket. [2]

The wave has a frequency of (**7.5 Hz / 7.5 cm/s / 7.5 cm**).

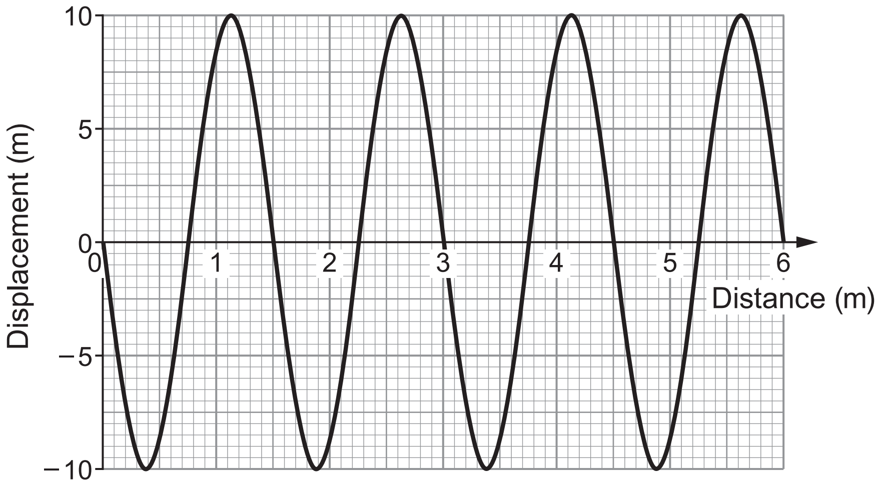
This means 7.5 waves pass a point each (**second / minute / hour**).

II. Use information from parts (b)(i) and (ii) to calculate the wave speed.
Use the equation:

wave speed = wavelength × frequency [2]

Wave speed = ... m/s

(c) The speed of the wave remains constant but its **frequency is doubled**.
A student draws a diagram of the expected wave. It is shown below.



Explain whether you agree with the student's diagram of the wave. [2]

....

....

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....

12

3420U101 07

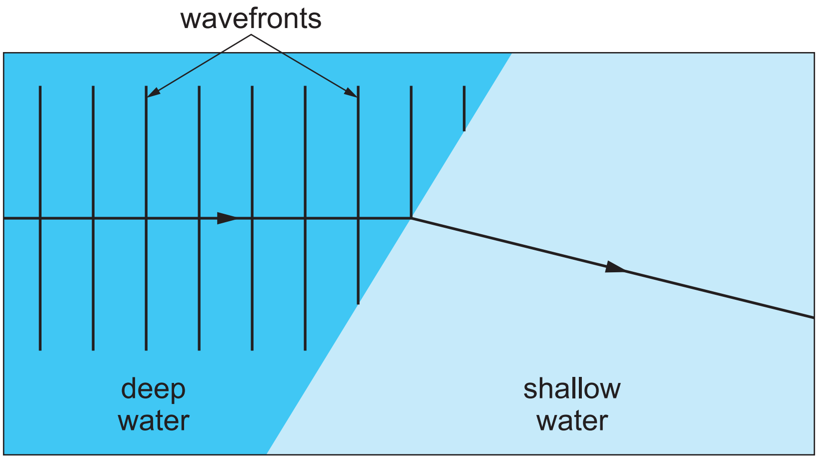
9. Water waves refract when they travel from deep water to shallow water.

Refraction happens because the speed of the wave changes.

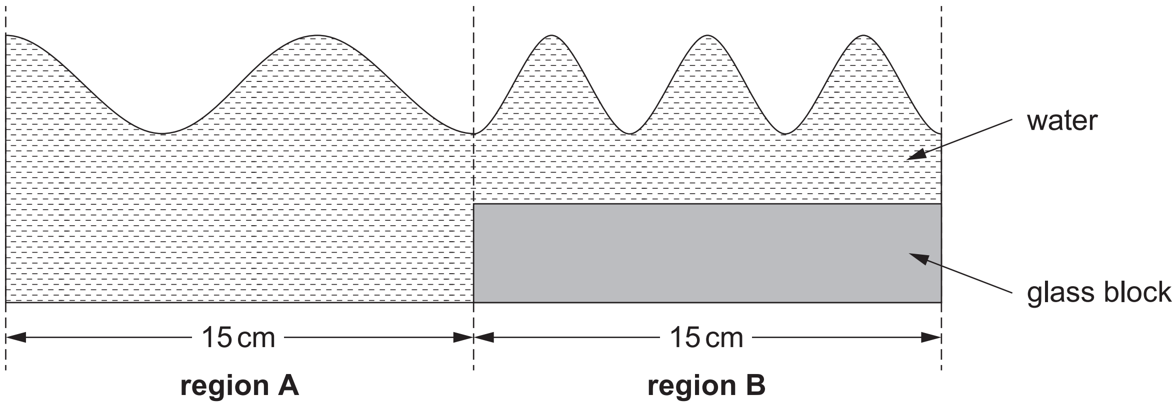
The wavelength also changes because the frequency is constant.

Students use a ripple tank to investigate this effect.

(a) **Complete the diagram below** to show the water waves in shallow water. [3]



(b) The depth of the shallow water can be changed by using glass blocks of different thicknesses.



Regions **A** and **B** are both 15 cm long.

(i) I. How many waves are shown in **region A**? .... [1]

II. Calculate the wavelength of the waves in **region A**. [1]

wavelength = ... cm

Examiner only

(ii) John says that the wave speed in **region B** is greater than the wave speed in **region A**. Explain whether John is correct. [2]

....

....

....

(c) In another experiment using a different tank, students investigate how the depth of water affects wave speed.

They change the depth of the water using different thickness glass blocks.

The water level is kept constant at 10 cm.

The table below shows their results.

Thickness of glass block (cm)	Depth of water (cm)	Wave speed (cm/s)
8	2	60
6	4	75
4	....	82

(i) **Complete the table.** [1]

(ii) Use the equation:

$$\text{wavelength} = \frac{\text{wave speed}}{\text{frequency}}$$

to calculate the wavelength of water waves of frequency 50 Hz when the thickness of the glass block is 6 cm. [2]

wavelength = ... cm

TURN OVER FOR THE LAST PART OF THE QUESTION

Examiner only

(iii) Janet states that when the thickness of the glass block decreases by 2 cm the wave speed increases by a quarter.
Explain to what extent Janet is correct. [3]
Space for calculations.

....

....

13

END OF PAPER